

MODULE 2: MATERIALS MANAGEMENT

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Material Management

Material management refers to the systematic planning, sourcing, purchasing, moving, storing, and controlling of materials required in a manufacturing process. The goal is to ensure the right quality and quantity of materials are available at the right time, at the right place, and at the lowest possible cost.

1. Functions of Material Management

- i. **Planning:** Determining the material requirements in advance to ensure the smooth operation of the production process.
- ii. **Purchasing:** Procuring materials from suppliers in accordance with pre-established quality standards, delivery schedules, and costs.
- iii. **Inventory Control:** Managing inventory levels to avoid overstocking and stockouts, reducing holding costs while ensuring uninterrupted production.
- iv. **Receiving and Inspection:** Ensuring that received materials meet the required specifications and quality.
- v. **Storage:** Organizing and storing materials in warehouses or other facilities, keeping them safe from damage or deterioration.
- vi. **Issuing Materials:** Delivering materials to the production department or other relevant departments as per the production schedule.
- vii. **Waste Management:** Minimizing material wastage by efficiently using resources and recycling where possible.

2. Importance of Material Management

- i. **Cost Control:** Proper material management helps in minimizing costs associated with material procurement, storage, and wastage.
- ii. **Operational Efficiency:** Ensures the smooth flow of materials to avoid production delays caused by material shortages or disruptions.
- iii. **Inventory Optimization:** Helps maintain optimal stock levels, avoiding both overstock and stockouts.
- iv. **Quality Assurance:** Guarantees that the correct materials of the right quality are used in production, leading to better product quality.
- v. **Supplier Relationships:** Establishing reliable supplier networks ensures timely delivery and cost-effective procurement of materials.

- vi. **Waste Reduction:** Reduces wastage through efficient usage, recycling, and timely disposition of unused materials.

3. Relationship with Other Departments

- i. **Production Department:** Material management ensures that the production department gets the materials it needs at the right time, avoiding disruptions in production.
- ii. **Finance Department:** Works closely with the finance department to control costs, approve purchase budgets, and manage cash flow related to procurement.
- iii. **Quality Control Department:** Coordinates with quality control to ensure that materials meet the required standards before they are accepted and used in production.
- iv. **Sales and Marketing:** Ensures sufficient inventory levels to meet customer demand and supports the sales department in meeting delivery commitments.
- v. **Logistics/Transportation:** Collaborates with logistics to ensure timely transportation and distribution of materials between warehouses and production sites.

Purchase

1. Objectives of Purchase:

The purchasing function aims to procure materials and services needed for production while balancing cost, quality, and time. Key objectives include:

- i. **Right Quality:** Ensure that materials meet the required specifications and standards.
- ii. **Right Quantity:** Procure the exact quantity required to avoid shortages or excessive inventory.
- iii. **Right Time:** Ensure timely delivery of materials to avoid production delays.
- iv. **Right Source:** Select reliable suppliers to ensure consistency in quality, price, and delivery.
- v. **Right Price:** Obtain materials at the most economical price, balancing cost-effectiveness with quality.
- vi. **Right Place:** Ensure that materials are delivered to the correct location, facilitating smooth operations.

2. Purchasing Systems:

Different organizations may adopt various purchasing systems based on their size, nature, and specific needs:

- i. **Centralized Purchasing:** All purchasing activities are handled by a single department or location, ensuring better control over purchasing decisions and economies of scale.
- ii. **Decentralized Purchasing:** Each department or division manages its own purchases, providing greater flexibility and quicker decision-making.
- iii. **Just-in-Time (JIT) Purchasing:** Materials are purchased only as they are needed in the production process, reducing inventory costs.
- iv. **Consignment Purchasing:** The supplier maintains ownership of the materials until they are used by the company. Payment is made only for the consumed materials.
- v. **Rate Contract Purchasing:** Long-term agreements are made with suppliers for regular supply at a predetermined price, ensuring stability in material costs.

3. Purchase Procedure:

A typical purchase procedure involves several steps to ensure efficiency and compliance:

- i. **Identifying the Requirement:** The department requiring materials or services raises a purchase requisition.
- ii. **Selecting Suppliers:** Potential suppliers are identified, and quotations are requested.
- iii. **Negotiation:** Prices, terms of delivery, and payment conditions are negotiated with suppliers.

- iv. **Issuing Purchase Orders:** Once a supplier is selected, a purchase order (PO) is issued, formally confirming the order.
- v. **Receiving and Inspecting Materials:** Upon delivery, materials are inspected for quality and quantity to ensure they match the order specifications.
- vi. **Payment Processing:** Once the materials are accepted, the purchase department processes the invoice for payment according to agreed terms.
- vii. **Record Keeping:** Proper documentation is maintained, including purchase orders, invoices, and delivery receipts, for future reference and auditing.

4. Terms and Forms Used in Purchase Department:

- i. **Quotation:** A formal document from the supplier listing the price and terms for the requested goods or services.
- ii. **Purchase Requisition:** A form used by departments to request the purchase of goods or services, initiating the procurement process.
- iii. **Purchase Order (PO):** A legally binding document issued by the buyer to the supplier, specifying the materials, quantities, prices, and delivery terms.
- iv. **Invoice:** A bill issued by the supplier requesting payment for goods or services delivered.
- v. **Delivery Note:** A document provided by the supplier with the delivered goods, listing the items delivered and their quantities.
- vi. **Goods Received Note (GRN):** A form used to record the receipt of goods, confirming that they have been received and inspected.
- vii. **Lead Time:** The time taken from placing the purchase order to receiving the materials.
- viii. **Terms of Payment:** The conditions agreed upon for payment, such as payment within 30 days (net 30) or upon delivery (cash on delivery).
- ix. **Bill of Materials (BOM):** A detailed list of the materials or components required for production, often used to guide the purchasing process.
- x. **Vendor:** Another term for a supplier; a business entity that provides goods or services to the company.

Store and Storekeeping

1. Storekeeping Functions: Storekeeping involves managing the storage, safety, and issuance of materials in a systematic manner. Key functions include:

- i. **Receipt of Materials:** Ensuring all incoming materials are inspected, verified, and accepted according to the purchase order.
- ii. **Storage of Materials:** Safely storing materials in a manner that preserves their quality and ensures easy access for future use.
- iii. **Inventory Control:** Monitoring stock levels to avoid overstocking or stockouts and to maintain an optimal level of inventory.
- iv. **Issuing Materials:** Providing materials to the production or other departments as per their needs and requisitions.
- v. **Safety and Security:** Protecting materials from theft, damage, or deterioration by ensuring proper storage conditions and security measures.
- vi. **Record Keeping:** Maintaining detailed records of materials received, issued, and stored to ensure traceability and accurate reporting.
- vii. **Stock Auditing:** Periodic checks of inventory to ensure that physical stock matches the recorded quantities.
- viii. **Disposal of Obsolete or Scrap Materials:** Managing the removal and proper disposal of materials that are no longer usable or are damaged.

- ix. **Security of Materials:** The store ensures the safety of materials through proper storage techniques and security measures to prevent theft or damage.

2. Classification of Stores

A. Centralized Stores: In a centralized system, all materials and supplies are stored in one main location, from which they are distributed to various departments or units as needed.

Advantages of Centralized Stores:

- **Better Control:** Centralized control over inventory helps in maintaining uniform policies and procedures.
- **Economies of Scale:** Bulk purchasing reduces costs due to discounts and better negotiation with suppliers.
- **Reduction in Redundancy:** Avoids duplicate inventory across departments, leading to cost savings.
- **Easy Supervision:** Centralized locations make it easier to manage, supervise, and audit the store's operations.
- **Efficient Use of Space:** Consolidates storage needs, reducing the overall space required for storing materials.

Disadvantages of Centralized Stores:

- **Delay in Material Distribution:** Since materials need to be transported from one location to different departments, it may cause delays.
- **Overloading:** Centralized stores may become overburdened with the responsibility of serving multiple departments.
- **Logistical Costs:** The cost of transporting materials from a central location to various departments may increase.
- **Limited Responsiveness:** Centralized stores may be less flexible in handling urgent material requirements in individual departments.

Applications of Centralized Stores:

- Large-scale industries where control, efficiency, and cost savings are more important than responsiveness.
- Organizations with standardized operations, such as manufacturing units, where all materials are required in bulk at set intervals.

B. Decentralized Stores: In a decentralized system, materials are stored in multiple locations close to the point of use. Each department or unit may have its own store.

Advantages of Decentralized Stores:

- **Quick Access to Materials:** Departments can quickly access materials from their own stores, reducing waiting times.
- **Greater Flexibility:** Each department has more control over its own inventory, allowing for quicker decision-making.
- **Reduced Transportation Costs:** Materials are stored close to where they are needed, reducing the need for transportation.
- **Customized Storage:** Different departments can manage their own materials based on specific needs.

Disadvantages of Decentralized Stores:

- **Higher Inventory Costs:** Each department must maintain its own inventory, leading to potential overstocking or duplication of items.
- **Less Control:** It is harder to maintain centralized control over inventory levels, leading to discrepancies and inefficiencies.
- **Increased Management Complexity:** Managing multiple stores in different locations requires more administrative effort and resources.
- **Loss of Economies of Scale:** Smaller, separate purchases may result in higher costs compared to bulk purchasing in centralized systems.

Applications of Decentralized Stores:

- Industries with diverse departments or operations requiring highly specific materials, such as hospitals, universities, or large organizations with distinct operational units.
- Companies where responsiveness and quick access to materials are critical, like in the maintenance departments of plants or repair facilities.

3. Application in Actual Practice: In actual practice, the choice between centralized and decentralized stores depends on the nature of the business and its operational needs:

- **Centralized Stores** are typically used by large manufacturing organizations, where uniformity in materials and bulk purchasing can reduce costs and enhance control. These stores are more common in industries where operations are predictable, and materials can be scheduled and distributed efficiently.
- **Decentralized Stores** are more suited to industries or organizations where quick access to materials is crucial, such as service industries, healthcare, and retail. Decentralized stores are also found in companies with geographically dispersed units or diverse product lines, where each unit has its own specific material needs.

4. Types of Records Maintained by Store: Stores maintain various records to manage inventory, track material flow, and ensure accountability:

- i. **Bin Card:** A physical record attached to storage bins that tracks material movements (receipts, issues, and balances) for each item stored.
- ii. **Store Ledger:** A centralized record, often maintained digitally, that keeps track of all stock transactions, including quantities, dates, and costs of materials.
- iii. **Goods Received Note (GRN):** A document used to confirm that the received goods match the purchase order in terms of quantity and quality.
- iv. **Material Requisition Form:** A form filled out by departments requesting materials from the store, which authorizes the issue of materials.
- v. **Material Issue Note:** A record confirming that the requested materials have been issued to the department.
- vi. **Purchase Order File:** A record of all purchase orders issued to suppliers, including quantities, prices, and delivery terms.
- vii. **Inventory Report:** A periodic report showing the current stock levels, any discrepancies, and reordering requirements.
- viii. **Obsolete Material Report:** A report identifying materials that are no longer needed or have become obsolete and require disposal or liquidation.

5. Types and Applications of Storage Equipment: The type of storage equipment used in stores depends on the nature of the materials being stored, the available space, and the need for easy access and safety. Common types of storage equipment include:

- i. **Shelving:** Simple storage units used for smaller items that need easy access. They can be made of metal, wood, or plastic. **Application:** Suitable for lightweight materials, office supplies, or small tools.
- ii. **Pallet Racks:** Large steel or aluminium racks designed to store heavy materials or bulk items on pallets. **Application:** Ideal for warehouses or stores that handle heavy or bulky materials, like raw materials or finished goods.
- iii. **Bins and Containers:** Plastic or metal bins used for organizing small parts, tools, or components. **Application:** Used in assembly lines, workshops, or warehouses to keep small items organized.
- iv. **Cantilever Racks:** Racks with extended arms, designed for storing long items such as pipes, rods, or lumber. **Application:** Useful in industries dealing with long materials or structural items.
- v. **Cold Storage Units:** Refrigerated storage units used for materials that require controlled temperatures. **Application:** Used in the food industry, pharmaceutical industry, and any place where temperature-sensitive goods are stored.
- vi. **Automated Storage and Retrieval Systems (ASRS):** High-tech systems that automate the storage and retrieval of materials using robotic arms or conveyor systems. **Application:** Found in large-scale warehouses or industries with high-volume storage needs, such as distribution centres.

6. Need and General Methods for Codification of Stores:

A. Need for Codification: Codification of stores refers to assigning a unique code or number to each item in the inventory. This system is essential for efficient material handling and management. The primary reasons for codification include:

- i. **Simplifies Identification:** It eliminates confusion by assigning each item a distinct code, making identification easier.
- ii. **Standardization:** Ensures that similar items are classified and managed uniformly across the organization.
- iii. **Streamlines Processes:** Codification helps in streamlining purchase, storage, and issue processes, as items are easier to track.
- iv. **Prevents Duplication:** By clearly identifying items, codification avoids unnecessary duplication of materials.
- v. **Improves Inventory Control:** Makes it easier to monitor stock levels, reorder points, and usage patterns.
- vi. **Facilitates Communication:** Allows for better communication between departments, especially when materials are being requested or ordered.
- vii. **Aids in Auditing:** Simplifies inventory audits, as materials can be tracked using their assigned codes.

B. General Methods for Codification: There are several methods used to codify materials in stores. The choice of method depends on the nature of the business and the complexity of the inventory:

- i. **Numeric Coding:** Assigns a unique number to each item, often organized by categories. **Example:** 1001 for raw materials, 2001 for finished products.
- ii. **Alphabetic Coding:** Uses letters to represent items or categories of materials. **Example:** RM for raw materials, FG for finished goods.
- iii. **Alpha-Numeric Coding:** Combines letters and numbers to provide more detailed coding, often used in complex systems with many different categories. **Example:** RM100 for raw material 100, FG500 for finished goods 500.
- iv. **Colour Coding:** Different colours are used to categorize items, typically for quick visual identification. **Example:** Red labels for perishable items, green labels for non-perishable.
- v. **Barcoding:** A more advanced method where each item is assigned a barcode that can be scanned for easy tracking. **Example:** Used in retail stores or large warehouses for fast identification and stock control.
- vi. **Hierarchical Coding:** Items are grouped into categories and subcategories, with codes reflecting the hierarchical structure. **Example:** 01-05-100, where 01 represents the category, 05 the subcategory, and 100 the specific item.

Inventory Control

Inventory control is the process of managing an organization's stock of materials and goods to ensure that the right quantity of items is available when needed while minimizing costs associated with holding and purchasing inventory. It involves tracking inventory levels, managing the flow of materials, and ensuring a balance between supply and demand to avoid overstocking or stockouts.

Objectives of Inventory Control

- i. **Ensuring Continuous Supply:** To ensure that the required materials or products are available when needed to avoid production delays or stockouts.
- ii. **Minimizing Holding Costs:** To reduce the cost of storing inventory, including warehousing, insurance, and depreciation.
- iii. **Minimizing Ordering Costs:** To reduce the costs associated with ordering, receiving, and processing inventory.
- iv. **Optimizing Stock Levels:** To maintain an optimal balance between stock quantity and the costs of ordering and holding inventory.
- v. **Reducing Risk of Obsolescence:** To avoid overstocking items that may become obsolete or expire.
- vi. **Maintaining Quality:** To ensure that materials and products in stock are of the right quality and are not damaged or deteriorated.
- vii. **Efficient Use of Capital:** To ensure that working capital is not tied up unnecessarily in excess inventory, which could be used for other productive purposes.
- viii. **Improving Customer Service:** To meet customer demands promptly by ensuring the availability of finished products.

Economic Order Quantity (EOQ)

The **Economic Order Quantity (EOQ)** is a formula used to determine the optimal order quantity that minimizes the total inventory costs, which include ordering costs and holding costs. **Key Parameters:**

- **D:** Annual demand for the product (units/year).
- **S:** Cost of placing an order (ordering cost per order).
- **H:** Holding cost per unit per year (cost of holding one unit in inventory for a year).

- **Q**: order quantity.

Total Cost of Inventory (TC): The total cost of inventory includes two components:

1. **Ordering Cost:** The cost associated with placing orders for inventory. The total ordering cost is calculated as:
2. **Holding Cost:** The cost of holding or storing inventory over a period. The total holding cost is calculated as:

$$\text{Ordering Cost} = \frac{D}{Q} \times S$$

$$\text{Holding Cost} = \frac{Q}{2} \times H$$

Total Cost (TC) = Ordering Cost + Holding Cost:

$$TC = \frac{D}{Q} \times S + \frac{Q}{2} \times H$$

To find the EOQ (optimal order quantity), we minimize the total cost function by differentiating the total cost with respect to Q and setting it equal to zero:

$$\begin{aligned} \frac{d(TC)}{dQ} &= -\frac{D}{Q^2} \times S + \frac{H}{2} = 0 \\ &= \frac{D}{Q^2} \times S = \frac{H}{2} \\ &= Q^2 = \frac{2DS}{H} \\ &= Q = \sqrt{\frac{2DS}{H}} = EOQ \end{aligned}$$

Key Insights from the EOQ Formula:

- EOQ gives the ideal order quantity that minimizes the total cost.
- Higher demand or ordering costs will result in larger EOQ.
- Higher holding costs will lead to smaller EOQ, as holding inventory becomes more expensive.

Inventory Analysis Methods

1. ABC Analysis: ABC analysis is a method used to categorize inventory items into three classes (A, B, and C) based on their importance, typically by annual consumption value (usage). This helps in prioritizing control and management efforts on the most valuable items. ABC analysis helps in focusing efforts and resources on managing the most critical items in the inventory (A items), while less focus is given to lower-value items (C items).

- **A items:** These are high-value items that account for a small percentage of the total inventory but a significant portion of the total value. Close monitoring and tight control are required. Example: Expensive machinery parts.
- **B items:** These are moderate-value items that account for a larger percentage of inventory but less value compared to A items. They require less strict control. Example: Regular office supplies.

- **C items:** These are low-value items that make up a large percentage of inventory but a small percentage of the total value. These items are controlled less strictly. Example: Stationery or screws.

Other Modern Methods of Analysis:

- **VED Analysis (Vital, Essential, Desirable):** Classifies items based on their importance to the production process. Vital items are essential for operations, while desirable items are of least importance.
- **HML Analysis (High, Medium, Low):** Categorizes items based on their unit price or cost. High-cost items require more control compared to low-cost items.
- **FSN Analysis (Fast, Slow, Non-moving):** Categorizes inventory based on the frequency of use. Fast-moving items are consumed quickly, while non-moving items may require liquidation or special handling.
- **XYZ Analysis:** Based on the predictability of demand, where X items have a stable demand, Y items have moderate fluctuations, and Z items have irregular demand.

Various Types of Inventory Models:

- i. **Wilson's Inventory Model (EOQ Model):** This is the most basic and widely used inventory model, which determines the optimal order quantity (EOQ) to minimize total inventory costs, including ordering and holding costs.
 - The Wilson model is based on the EOQ formula, where the goal is to find the balance between ordering frequently (leading to lower holding costs) and ordering infrequently (leading to higher holding costs). This model helps organizations in calculating the optimal order quantity to minimize costs.
- ii. **Replenishment Model:** In this model, inventory is replenished whenever it reaches a predetermined level known as the reorder point. The reorder quantity could be based on EOQ or a fixed amount.
 - The organization sets a fixed inventory level, and when stock falls to this level, an order is placed. The replenishment is done either periodically or continuously based on usage patterns. The replenishment model ensures that stock is available just in time to meet demand without holding excessive inventory.
- iii. **Two-bin Model:** In this simple inventory control system, two bins are used to store items. The first bin holds the working stock, and the second bin holds the reserve stock.
 - When the first bin is emptied, an order is placed for more stock, and the second bin's stock is used until the new stock arrives. The two-bin model is commonly used for small, inexpensive items. This model is practical and easy to implement for items that are consumed regularly and where the cost of stockouts is high, but items themselves are inexpensive.

Material Requirement Planning (MRP)

Material Requirement Planning (MRP) is a production planning and inventory control system used to manage manufacturing processes. MRP ensures that materials and components are available for production, that products are available for delivery to customers, and that inventory levels are maintained at optimal levels. It helps organizations efficiently manage the procurement, production, and delivery of materials and goods.

MRP is based on the **Bill of Materials (BOM)**, which lists the components required to manufacture a product, and it uses production schedules and inventory data to calculate the materials needed for production. **Key Components of MRP:**

- i. **Master Production Schedule (MPS):** A detailed plan that specifies when and how much of each product should be produced.
- ii. **Bill of Materials (BOM):** A structured list of all the raw materials, components, and subassemblies needed to produce a finished product.
- iii. **Inventory Records:** Data on current stock levels, outstanding orders, and lead times for materials.
- iv. **Lead Times:** The time required to order and receive materials, which MRP considers for scheduling procurement.

Applications of MRP:

- i. **Production Scheduling:** MRP helps in scheduling production processes by ensuring the availability of raw materials at the right time and in the correct quantity.
- ii. **Inventory Management:** It optimizes inventory levels, reducing excess stock and preventing shortages by planning materials according to demand forecasts.
- iii. **Procurement Planning:** MRP helps procurement teams plan when and how much to order based on production needs, lead times, and available inventory.
- iv. **Demand Forecasting:** By analysing production schedules and customer orders, MRP can forecast the demand for raw materials and finished products.
- v. **Cost Control:** It reduces carrying costs by maintaining optimal inventory levels and helps avoid rush orders, reducing procurement costs.
- vi. **Resource Allocation:** Helps allocate resources such as labour and machinery more efficiently by providing accurate material availability data.
- vii. **Improved Customer Satisfaction:** Ensures timely production and delivery of goods by ensuring materials are available when needed.

MRP Process Flow:

- i. **Demand Inputs:** Customer orders, forecasts, or market demand drive the MRP process.
- ii. **Master Production Schedule (MPS):** MPS defines what needs to be produced and when.
- iii. **MRP Calculation:** Based on the MPS, the BOM, inventory records, and lead times, MRP calculates what materials need to be ordered or produced.
- iv. **Planned Orders:** The system generates planned orders for purchasing or production.
- v. **Order Release:** Orders are released to suppliers or production departments.
- vi. **Order Tracking:** MRP tracks open orders and adjusts based on changes in demand or supply conditions.

Software Packages for MRP: Several software solutions help companies implement MRP. These tools offer features such as production scheduling, inventory management, and procurement planning.

Popular MRP Software Packages:

1. **SAP ERP:** SAP offers a robust and comprehensive ERP solution that includes MRP as a key module. It is widely used by large organizations for its ability to integrate MRP with other functions such as finance, HR, and sales.
 - **Key Features:** Advanced demand forecasting, detailed inventory control, and integrated supply chain management.

- **Target Users:** Large enterprises across industries like manufacturing, automotive, and retail.
2. **Oracle NetSuite:** NetSuite is a cloud-based ERP solution that provides a full suite of business management tools, including MRP. It helps businesses optimize production planning, inventory management, and procurement.
 - **Key Features:** Real-time inventory tracking, demand planning, and integration with financial systems.
 - **Target Users:** Small to mid-sized businesses and growing enterprises.
 3. **Microsoft Dynamics 365:** Microsoft Dynamics offers a flexible and scalable MRP solution as part of its enterprise resource planning (ERP) suite. It is designed to handle complex manufacturing requirements.
 - **Key Features:** MRP with production scheduling, demand forecasting, and supplier management.
 - **Target Users:** Medium to large enterprises in manufacturing, retail, and distribution.
 4. **Odoo:** Odoo is an open-source ERP that offers MRP as part of its extensive module library. It is a flexible, cost-effective solution for small to medium-sized companies.
 - **Key Features:** MRP with inventory tracking, procurement planning, and multi-level BOM management.
 - **Target Users:** Small and medium enterprises (SMEs) looking for an affordable, customizable MRP solution.
 5. **Fishbowl Inventory:** Fishbowl is an inventory management software that also offers MRP functionality for small to mid-sized businesses. It integrates with QuickBooks for financial management.
 - **Key Features:** Inventory control, production scheduling, and BOM management.
 - **Target Users:** SMEs and businesses using QuickBooks for accounting.
 6. **DELMIAworks (formerly IQMS):** DELMIAworks is an ERP and manufacturing software that includes MRP functionality, focusing on optimizing manufacturing processes and resource planning.
 - **Key Features:** Real-time production tracking, detailed MRP analytics, and inventory optimization.
 - **Target Users:** Manufacturing businesses of all sizes.